

1	(a)	(i)	Distance = area under the $v - t$ graph = $\frac{1}{2}(8.8)(0.90) = 4.0 \text{ m}$	B1
		(ii)	average acceleration = $\frac{\text{change in velocity}}{\text{time}}$ $= \frac{(-4.4 - 8.8)}{0.50}$ $= -26 \text{ m s}^{-2}$ <i>(Circle but not penalise for missing -ve sign)</i>	C1 A1
	(b)	(i)	The accelerations are the same as <u>the gradients of the graph before and after the rebound are the same</u> . (the girl is in a state of free fall before and after the rebound).	B1
		(ii)	The area under the lines represents height and the <u>second area is smaller</u> than the first area, hence, <u>the rebound height is less</u> than the initial height. Thus the <u>GPE of the girl at $t = 0$ is larger than the GPE at $t = 1.85 \text{ s}$</u>. OR The kinetic energy (KE) of the girl at trampoline equals to the GPE at maximum height. The speed after rebound is smaller and hence the <u>KE becomes smaller</u>. Thus the gravitational potential of the girl after rebound is less.	M1 A1 (M1) (A1)

2	(a)		Using hinge as pivot:	
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